PRODUCT SPECIFICATION MANUAL

Dredging Equipment Series

Marine Engineering Division

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Prepared for: Benamdaj Dredging Solutions Ltd.

Classification: Confidential — Commercial in Confidence

Document Control

1.1 Document Information

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Models Covered	16/14", 14/12", 12/10", 10/10" Dredger Series (selected models — additional fleet vessels exist)

1.2 Revision History

Version	Date	Author	Description
1.0	April 2025	[Author]	Initial document creation — compiled from company data

1.3 Purpose & Scope

This Product Specification Manual provides the authoritative technical reference for four selected dredger models from the Benamdaj Dredging Solutions Ltd. fleet. It serves customer support, pre-sales engineering, maintenance planning and spare parts identification.

1.4 Models Covered

Pump Size	Model Name	Drive System	Engine
16/14 Inch	Cutter Suction Dredger	Marine gearbox	Weichai 6170 (650 HP) + CAT 3412 Aux.
14/12 Inch	Amphibious Multifunctional Dredger	Marine gearbox	Weichai 6170 (650 HP) + CAT 3412 Aux.
12/10 Inch	Bucket Chain Dredger	Marine gearbox	CAT 3408 Main + CAT 3412 Aux.
10/10 Inch	Jet Suction Dredger	PTO drive	Weichai WP13 (450 HP) + CAT 3412 Aux.

1.5 Definitions & Abbreviations

Term	Definition
CSD	Cutter Suction Dredger
PTO	Power Take-Off — drive system on 10/10 Inch model
kW / HP	Kilowatt / Horsepower
m^3/hr	Cubic metres per hour
IMO	International Maritime Organization
Aux.	Auxiliary engine (CAT 3412, all models)
NTS	Not To Scale

Product Specification: 16/14 Inch Cutter Suction Dredger

The 16/14 Inch Cutter Suction Dredger is the highest-capacity vessel in this documented series, designed for high-volume dredging in rivers, harbours, and coastal waterways. Powered by a Weichai 6170 series marine diesel engine transmitting power to the dredge pump via a marine gearbox, with a Caterpillar 3412 auxiliary engine for onboard electrical systems. The vessel uses a jet suction system and is positioned by a manual anchor winch arrangement with 4 anchor points.

1. General Information

GENERAL INFORMATION	
Model Name	**Cutter Suction Dredger**
Model Code / Part No.	**BDS-CSD-1614**
Product Family	**Cutter Suction Dredger Series**
Classification	**Cutter Suction Dredger (CSD)**
Pump Configuration	**16-inch suction / 14-inch discharge**
Primary Application	**High-volume river dredging, harbour maintenance, land reclamation**
Country of Manufacture	**Nigeria**

2. Engine & Power System

MAIN ENGINE	
Engine Manufacturer	**Weichai Power Co., Ltd**
Engine Model / Type	**X6170ZC650-2**
Power Output	**478 kW (650 HP)**
Number of Engines	**1**
Fuel Type	**Diesel**
Engine Speed (rated)	**1,200 RPM**
Cooling System	**Freshwater cooled — turbocharged and aftercooled**
Emissions Standard	**IMO Tier II**

AUXILIARY ENGINE (ALL MODELS)	
Make / Model	**Caterpillar 3412**
Purpose	**Onboard electrical generation and auxiliary systems**
Fuel Type	**Diesel**

DRIVE SYSTEM — MARINE GEARBOX	
Drive System Type	**Marine Gearbox**
Gearbox Ratio (engine:pump)	**2:1**

4. Dredging Pump System

DREDGING PUMP SYSTEM	
Pump Type	**Single-stage centrifugal slurry pump**
Pump Inlet Diameter	**400 mm (16 in)**
Pump Outlet Diameter	**350 mm (14 in)**
Impeller Diameter	**1,016 mm (confirm variant with company)**
Max Flow Rate (water)	**720 – 3,600 m³/hr**
Max Flow Rate (slurry)	**2,500 – 6,000 m³/hr**
Pump Shaft Speed	**300 – 500 RPM**
Pump Drive Type	**Marine gearbox (engine to pump)**
Discharge Pipe Diameter	**16 inches (406 mm)**
Max Discharge Distance	**1,000 m**
Max Dredging Depth	**22 m**
Pump Casing Material	**High-chrome alloy (Cr27/Cr30)**
Impeller Material	**High-chrome alloy (Cr27/Cr30)**

5. Cutter Head & Ladder System

CUTTER HEAD & LADDER SYSTEM	
Cutter Head Type	**Jet suction system**
Cutter Drive Type	**Manual — winch operated**
Ladder Type	**Up ladder**
Ladder Lift System	**Manual winch**

6. Operational Capabilities

OPERATIONAL CAPABILITIES	
Max Production Rate (bulk)	**500 m³/hr**
Soil Types — suitable	**Normal muddy soil, soft sediment, sand, silt**
Max Operating Water Depth	**6.5 m**
Positioning System	**Manual anchor winch system**
Number of Anchors / Winches	**4 anchors**

OPERATIONAL CAPABILITIES	
Spud Type (if fitted)	**None fitted**
Propulsion Type	**Non-self-propelled — repositioned via anchor winch**

7. Electrical & Control Systems

Note: Electrical and control system configuration is identical across all four models.

ELECTRICAL & CONTROL SYSTEMS	
Onboard Generator Capacity	**15 kVA (Caterpillar 3412 auxiliary engine)**
Operating Voltage	**240 V**
Frequency	**50 Hz**
Control System Type	**Manual operator controls**
HMI Type	**N/A — manual gauges and levers only**
Automation Level	**Fully manual**
GPS / Dredge Management System	**None — manually positioned**
Communication Systems	**Radio communication**
Additional Electrical Equipment	**Solar panels / welding machines**
PLC	**N/A — no PLC fitted**

8. Product Images & Diagrams

9. Maintenance & Service

Note: Maintenance schedule and service requirements are identical across all four models.

MAINTENANCE & SERVICE SCHEDULE	
Engine Service Interval	**Every 200 hours**
Pump Wear Inspection Interval	**Every 200 hours**
Cutter Bearing Grease Interval	**Every 200 hours**
Hydraulic Oil Change Interval	**Every 200 hours**
Hull Survey Interval	**Annual (every 12 months)**
Recommended Engine Oil	**Total Ruby X**
Recommended Cutter Bearing Grease	**Weekly basis (confirm grade with supplier)**
Recommended Hydraulic Fluid	**Weekly check basis (confirm grade with supplier)**
Total Grease Points on Vessel	**7**
Special Tools Required	**Complete set of spanners**

Product Specification: 14/12 Inch Amphibious Multifunctional Dredger

The 14/12 Inch Amphibious Multifunctional Dredger combines dredging capability with amphibious mobility, enabling operation in shallow water, marshes, and areas inaccessible to conventional hull-mounted dredgers. Powered by a Weichai 6170 series engine and marine gearbox, it achieves a maximum in-situ production rate of 300 m³/hr.

1. General Information

GENERAL INFORMATION	
Model Name	**Amphibious Multifunctional Dredger**
Model Code / Part No.	**BDS-AMD-1412**
Product Family	**Amphibious Dredger Series**
Classification	**Amphibious Multifunctional Dredger**
Pump Configuration	**14-inch suction / 12-inch discharge**
Primary Application	**Amphibious dredging in shallow waterways, wetlands, and confined riverine environments**
Country of Manufacture	**Nigeria**

2. Engine & Power System

MAIN ENGINE	
Engine Manufacturer	**Weichai Power Co., Ltd**
Engine Model / Type	**X6170ZC650-2**
Power Output	**478 kW (650 HP)**
Number of Engines	**1**
Fuel Type	**Diesel**
Engine Speed (rated)	**1,200 RPM**
Cooling System	**Freshwater cooled — turbocharged and aftercooled**
Emissions Standard	**IMO Tier II**

AUXILIARY ENGINE (ALL MODELS)	
Make / Model	**Caterpillar 3412**
Purpose	**Onboard electrical generation and auxiliary systems**
Fuel Type	**Diesel**

DRIVE SYSTEM — MARINE GEARBOX	
Drive System Type	**Marine Gearbox**
Gearbox Ratio (engine:pump)	**2:1**

4. Dredging Pump System

DREDGING PUMP SYSTEM	
Pump Type	**Single-stage centrifugal slurry pump**
Pump Inlet Diameter	**350 mm (14 in)**
Pump Outlet Diameter	**300 mm (12 in)**
Impeller Diameter	**864 mm**
Max Flow Rate (water)	**576 – 3,024 m^3/hr**
Max Flow Rate (slurry)	**3,000 m^3/hr**
Pump Shaft Speed	**300 – 700 RPM**
Pump Drive Type	**Marine gearbox (engine to pump)**
Discharge Pipe Diameter	**14 inches (356 mm)**
Max Discharge Distance	**600 m**
Max Dredging Depth	**22 m**
Pump Casing Material	**High-chrome alloy (Cr27/Cr30)**
Impeller Material	**High-chrome alloy (Cr27/Cr30)**

5. Cutter Head & Ladder System

CUTTER HEAD & LADDER SYSTEM	
Cutter Head Type	**Jet Suction System**
Cutter Drive Type	**Manual — winch operated**
Ladder Type	**Up Ladder**
Ladder Lift System	**Manual winch**

6. Operational Capabilities

OPERATIONAL CAPABILITIES	
Max Production Rate (in situ)	**300 m^3/hr**
Soil Types — suitable	**Normal muddy soil, soft sediment, sand, silt**
Positioning System	**Manual anchor winch system**
Propulsion Type	**Non-self-propelled — repositioned via anchor winch**
Number of Anchors / Winches	**4 anchors**

7. Electrical & Control Systems

Note: Electrical and control system configuration is identical across all four models.

ELECTRICAL & CONTROL SYSTEMS	
Onboard Generator Capacity	**15 kVA (Caterpillar 3412 auxiliary engine)**
Operating Voltage	**240 V**
Frequency	**50 Hz**
Control System Type	**Manual operator controls**
HMI Type	**N/A — manual gauges and levers only**
Automation Level	**Fully manual**
GPS / Dredge Management System	**None — manually positioned**
Communication Systems	**Radio communication**
Additional Electrical Equipment	**Solar panels / welding machines**
PLC	**N/A — no PLC fitted**

8. Product Images & Diagrams

9. Maintenance & Service

Note: Maintenance schedule and service requirements are identical across all four models.

MAINTENANCE & SERVICE SCHEDULE	
Engine Service Interval	**Every 200 hours**
Pump Wear Inspection Interval	**Every 200 hours**

MAINTENANCE & SERVICE SCHEDULE	
Cutter Bearing Grease Interval	**Every 200 hours**
Hydraulic Oil Change Interval	**Every 200 hours**
Hull Survey Interval	**Annual (every 12 months)**
Recommended Engine Oil	**Total Ruby X**
Recommended Cutter Bearing Grease	**Weekly basis (confirm grade with supplier)**
Recommended Hydraulic Fluid	**Weekly check basis (confirm grade with supplier)**
Total Grease Points on Vessel	**7**
Special Tools Required	**Complete set of spanners**

Product Specification: 12/10 Inch Bucket Chain Dredger

The 12/10 Inch Bucket Chain Dredger is the only model in the Benamdaj fleet powered by a Caterpillar 3408 main engine, driving the dredge pump via a marine gearbox. A Caterpillar 3412 auxiliary engine provides onboard electrical power, consistent with all other fleet models. Maximum in-situ production rate is 200 m³/hr.

1. General Information

GENERAL INFORMATION	
Model Name	**Bucket Chain Dredger**
Model Code / Part No.	**BDS-BCD-1210**
Product Family	**Bucket Chain Dredger Series**
Classification	**Bucket Chain Dredger (BCD)**
Pump Configuration	**12-inch suction / 10-inch discharge**
Primary Application	**River dredging, channel maintenance, moderate-depth sediment removal**
Country of Manufacture	**Nigeria**

2. Engine & Power System

MAIN ENGINE	
Engine Manufacturer	**Caterpillar Inc.**
Engine Model / Type	**3408**
Power Output	**520HP**
Number of Engines	**1**
Fuel Type	**Diesel**
Engine Speed (rated)	**1800 RPM**
Cooling System	**Freshwater cooled**
Emissions Standard	-

AUXILIARY ENGINE (ALL MODELS)	
Make / Model	**Caterpillar 3412**
Purpose	**Onboard electrical generation and auxiliary systems**
Fuel Type	**Diesel**

4. Dredging Pump System

DREDGING PUMP SYSTEM	
Pump Type	**Single-stage centrifugal slurry pump**
Pump Inlet Diameter	**300 mm (12 in)**
Pump Outlet Diameter	**250 mm (10 in)**
Impeller Diameter	**667 mm standard / 950 mm high-head (confirm with company)**
Max Flow Rate (water)	**360 – 1,440 m³/hr**
Max Flow Rate (slurry)	*To be confirmed*
Pump Shaft Speed	**350 – 700 RPM**
Pump Drive Type	**Marine gearbox (engine to pump)**
Discharge Pipe Diameter	**12 inches (305 mm)**
Max Discharge Distance	**22 m**
Max Dredging Depth	**600 m**
Pump Casing Material	**High-chrome alloy (Cr27/Cr30)**
Impeller Material	**High-chrome alloy (Cr27/Cr30)**

5. Cutter Head & Ladder System

CUTTER HEAD & LADDER SYSTEM	
Cutter Head Type	**Jet Suction System**
Cutter Drive Type	**Manual — winch operated**
Ladder Type	**Up Ladder**
Ladder Lift System	**Manual winch**

6. Operational Capabilities

OPERATIONAL CAPABILITIES	
Max Production Rate (in situ)	**200 m³/hr**
Soil Types — suitable	**Normal muddy soil, soft sediment, sand, silt**
Positioning System	**Manual anchor winch system**
Number of Anchors / Winches	**4 anchors**
Propulsion Type	**Non-self-propelled — repositioned via anchor winch**

7. Electrical & Control Systems

Note: Electrical and control system configuration is identical across all four models.

ELECTRICAL & CONTROL SYSTEMS	
Onboard Generator Capacity	**15 kVA (Caterpillar 3412 auxiliary engine)**
Operating Voltage	**240 V**
Frequency	**50 Hz**
Control System Type	**Manual operator controls**
HMI Type	**N/A — manual gauges and levers only**
Automation Level	**Fully manual**
GPS / Dredge Management System	**None — manually positioned**
Communication Systems	**Radio communication**
Additional Electrical Equipment	**Solar panels / welding machines**
PLC	**N/A — no PLC fitted**

8. Product Images & Diagrams

9. Maintenance & Service

Note: Maintenance schedule and service requirements are identical across all four models.

MAINTENANCE & SERVICE SCHEDULE	
Engine Service Interval	**Every 200 hours**
Pump Wear Inspection Interval	**Every 200 hours**
Cutter Bearing Grease Interval	**Every 200 hours**
Hydraulic Oil Change Interval	**Every 200 hours**
Hull Survey Interval	**Annual (every 12 months)**
Recommended Engine Oil	**Total Ruby X**
Recommended Cutter Bearing Grease	**Weekly basis (confirm grade with supplier)**
Recommended Hydraulic Fluid	**Weekly check basis (confirm grade with supplier)**

MAINTENANCE & SERVICE SCHEDULE	
Total Grease Points on Vessel	**7**
Special Tools Required	**Complete set of spanners**

Product Specification: 10/10 Inch Jet Suction Dredger

The 10/10 Inch Jet Suction Dredger is the most compact vessel in the Benamdaj fleet and the only model driven by a PTO (Power Take-Off) system rather than a marine gearbox. Powered by a Weichai WP13 marine diesel engine, it achieves a maximum dredging depth of 18 m and a discharge distance of 300 m. A Caterpillar 3412 auxiliary engine provides onboard electrical power, consistent with all other fleet models.

1. General Information

GENERAL INFORMATION	
Model Name	**Jet Suction Dredger**
Model Code / Part No.	**BDS-JSD-1010**
Product Family	**Jet Suction Dredger Series**
Classification	**Jet Suction Dredger (JSD)**
Pump Configuration	**10-inch suction / 10-inch discharge**
Primary Application	**Shallow water dredging, confined waterways, sediment removal**
Country of Manufacture	**Nigeria**

2. Engine & Power System

MAIN ENGINE	
Engine Manufacturer	**Weichai Power Co., Ltd**
Engine Model / Type	**WP13C550-18E121**
Power Output	**405 kW / 450 HP (nominal)**
Number of Engines	**1**
Fuel Type	**Diesel**
Engine Speed (rated)	**1,800 RPM**
Cooling System	**Water-cooled — turbocharged and aftercooled**
Emissions Standard	**IMO Tier II**

AUXILIARY ENGINE (ALL MODELS)	
Make / Model	**Caterpillar 3412**
Purpose	**Onboard electrical generation and auxiliary systems**
Fuel Type	**Diesel**

4. Dredging Pump System

DREDGING PUMP SYSTEM	
Pump Type	**Single-stage centrifugal slurry pump**
Pump Inlet Diameter	**250 mm (10 in)**
Pump Outlet Diameter	**250 mm (10 in)**
Impeller Diameter	**610 – 680 mm (confirm with company)**
Max Flow Rate (water)	**600 – 2,200 m³/hr**
Max Flow Rate (slurry)	**940 m³/hr typical**
Pump Shaft Speed	**500 – 950 RPM**
Pump Drive Type	**PTO drive (engine to pump)**
Discharge Pipe Diameter	**10 inches (254 mm)**
Max Discharge Distance	**300 m**
Max Dredging Depth	**18 m**
Pump Casing Material	**High-chrome alloy (Cr27/Cr30)**
Impeller Material	**High-chrome alloy (Cr27/Cr30)**

5. Cutter Head & Ladder System

CUTTER HEAD & LADDER SYSTEM	
Cutter Head Type	**Jet Suction System**
Cutter Drive Type	**Manual — winch operated**
Ladder Type	**Up Ladder**
Ladder Lift System	**Manual winch**

6. Operational Capabilities

OPERATIONAL CAPABILITIES	
Max Production Rate (bulk)	**100 m³/hr**
Soil Types — suitable	**Normal muddy soil, soft sediment, sand, silt**
Positioning System	**Manual anchor winch system**
Propulsion Type	**Non-self-propelled — repositioned via anchor winch**

7. Electrical & Control Systems

Note: Electrical and control system configuration is identical across all four models.

ELECTRICAL & CONTROL SYSTEMS	
Onboard Generator Capacity	**15 kVA (Caterpillar 3412 auxiliary engine)**
Operating Voltage	**240 V**

ELECTRICAL & CONTROL SYSTEMS	
Frequency	**50 Hz**
Control System Type	**Manual operator controls**
HMI Type	**N/A — manual gauges and levers only**
Automation Level	**Fully manual**
GPS / Dredge Management System	**None — manually positioned**
Communication Systems	**Radio communication**
Additional Electrical Equipment	**Solar panels / welding machines**
PLC	**N/A — no PLC fitted**

8. Product Images & Diagrams

9. Maintenance & Service

Note: Maintenance schedule and service requirements are identical across all four models.

MAINTENANCE & SERVICE SCHEDULE	
Engine Service Interval	**Every 200 hours**
Pump Wear Inspection Interval	**Every 200 hours**
Cutter Bearing Grease Interval	**Every 200 hours**
Hydraulic Oil Change Interval	**Every 200 hours**
Hull Survey Interval	**Annual (every 12 months)**
Recommended Engine Oil	**Total Ruby X**
Recommended Cutter Bearing Grease	**Weekly basis (confirm grade with supplier)**

MAINTENANCE & SERVICE SCHEDULE	
Recommended Hydraulic Fluid	**Weekly check basis (confirm grade with supplier)**
Total Grease Points on Vessel	**7**
Special Tools Required	**Complete set of spanners**

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